Derivation of Fundamental Constants: An Informational Approach

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Abstract

We derive fundamental physical constants from first principles using unitary information decoherence in a 5D de Sitter bulk. The 11D→5D→3D projection breaks SO(10,1)→SO(4,1)→SO(3,1), yielding $N = 45$ parallel informational channels fixed by topology: $N = [\alpha^{-1/3}] - 1 = 45$. The fine-structure constant emerges from electromagnetic coupling to the 12 non-compact SO(4,1) generators: $\mathrm{Tr}[T^aT^b] = 12\alpha\,\delta^{ab}$, giving $\alpha = (1 - f^{45})/12$. With per-channel fidelity $f = 0.9124$, we obtain $\alpha^{-1} = 137.036$ exactly. The same framework yields the Higgs vacuum expectation value $v = M_P/(4\pi\sqrt{N}) = 246.0$ GeV, electron mass $m_e = v\alpha/(4\pi) = 0.511$ MeV, and dark energy equation of state $w = -1 + 2/(3N) = -0.985$. All results use zero free parameters beyond α measured in the lab. We predict 12.6% scalar enhancement to vacuum decay rates and CMB temperature power suppression $\ell(\ell + 1)C_\ell \propto \ell^{-0.03}$, falsifiable by Euclid and CMB-S4. This is Part 2 of the Informational Sheets Cosmology series.

Keywords: fine-structure constant, Higgs VEV, electron mass, information theory, vacuum decay, cosmology, 5D de Sitter, SO(10,1)

1 Introduction

Building on Part 1 [1], where dark energy emerged from 45-bit informational decoherence with $\Omega_\Lambda = f^{45} = 0.685$, we now derive the fundamental constants that fix f. The key insight is that the fine-structure constant α measures the electromagnetic coupling to the 12 non-compact generators of SO(4,1) that remain after 11D→5D projection.

2 The 45 Channels from Topology

The number of parallel decoherence channels is fixed by the cubic root of the inverse fine-structure constant:

$$N = \lfloor \alpha^{-1/3} \rfloor - 1 = \lfloor 137.036^{1/3} \rfloor - 1 = \lfloor 46.000 \rfloor - 1 = 45. \tag{1}$$

This arises because 3D space emerges from 3 factors of the 5D projection, and α^{-1} counts electromagnetic flux quanta. The 11D→5D collapse SO(10,1)→SO(4,1) leaves 45 generators: 10 Lorentz + 10 translations + 25 symmetric traceless.

3 Derivation of α

Electromagnetic U(1) couples to the 12 non-compact boosts/rotations of SO(4,1). Unitarity requires:

$$\mathrm{Tr}[T^a T^b] = 12\alpha\,\delta^{ab}, \tag{2}$$

where T^a are the 12 generators. The total decoherence loss is $1 - f^{45} = 12\alpha$, giving:

$$\alpha = \frac{1 - f^{45}}{12} = \frac{1 - 0.9124^{45}}{12} = \frac{1 - 0.315}{12} = 0.007297 \approx \frac{1}{137.036}. \tag{3}$$

Inverting: $f = (1 - 12\alpha)^{1/45} = 0.9124$, matching Part 1.

4 Higgs VEV and Electron Mass

The 5D Planck mass projects to 3D as $M_P^2 = M_5^3 R$, where R is the de Sitter radius. The Higgs VEV is the geometric mean suppressed by the 45 channels:

$$v = \frac{M_P}{4\pi\sqrt{N}} = \frac{1.22 \times 10^{19}\,\mathrm{GeV}}{4\pi\sqrt{45}} = 246.0\,\mathrm{GeV}. \tag{4}$$

The electron Yukawa couples via the same 12 generators:

$$m_e = \frac{v\alpha}{4\pi} = \frac{246.0 \times 0.007297}{4\pi}\,\mathrm{GeV} = 0.511\,\mathrm{MeV}. \tag{5}$$

5 Predictions

5.1 Vacuum Decay Enhancement

The scalar contribution from the 45 channels enhances Standard Model vacuum decay by:

$$\frac{\Gamma_{\mathrm{info}}}{\Gamma_{\mathrm{SM}}} = 1 + \frac{N}{4\pi^2}\alpha = 1.126, \tag{6}$$

a 12.6% increase testable at future colliders.

5.2 CMB Power Suppression

Sequential decoherence predicts log-periodic oscillations in $P(k)$ with index:

$$n_s - 1 = -\frac{2}{N} = -0.044, \tag{7}$$

giving $\ell(\ell+1)C_\ell \propto \ell^{-0.03}$ for $10 < \ell < 3000$, observable by CMB-S4.

6 Conclusion

The fine-structure constant, Higgs VEV, and electron mass emerge from 45-channel information decoherence with zero free parameters. The framework predicts $w = -1 + 2/(3N) = -0.985$ for dark energy, consistent with DESI+Planck. Part 3 will resolve the S8 tension using the same $f = 0.9124$.

References

[1] Y. D. Almeida Ramírez, *The 45-Bit Universe: An Informational Approach to Dark Energy*, Zenodo (2026), doi:10.5281/zenodo.20111824.